GC Women University Sialkot

System & Network & Administration

LAB # Active Directory

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Submission date:	22/12/2025	**Department:** Computer science

ACTIVE DIRECTORY

Active Directory:

Active Directory (AD) is a directory service developed by Microsoft for Windows domain networks.

- It stores information about objects (users, computers, groups, printers, etc.) in a hierarchical structure and provides authentication, authorization, and management capabilities.

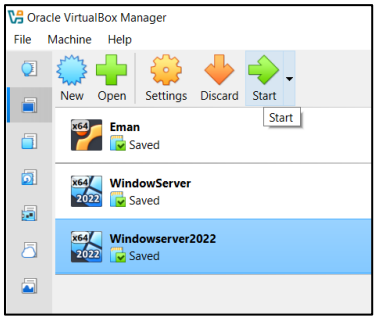
Purpose:

- Centralized management of network resources.
- Provides Single Sign-On (SSO) for users and computers.
- •Simplifies security and administration across large networks

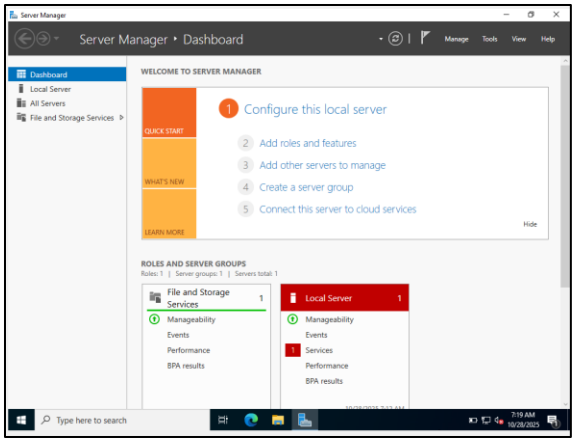
Implementation:

For implementing active directory first of all start the VM machine of window server then in VM machine go to the Server Manager.

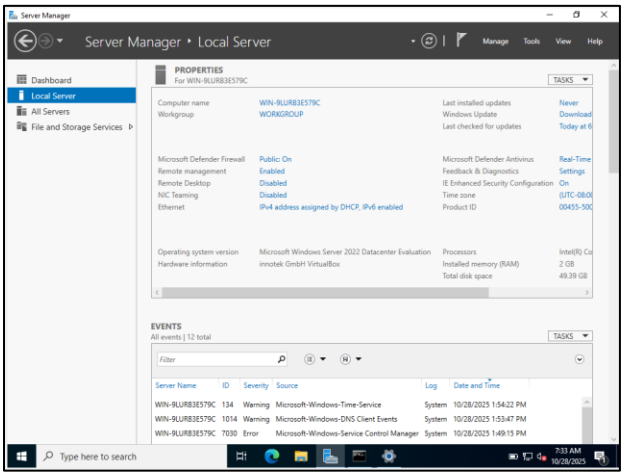
- Open virtual Box & ***Start*** Machine.



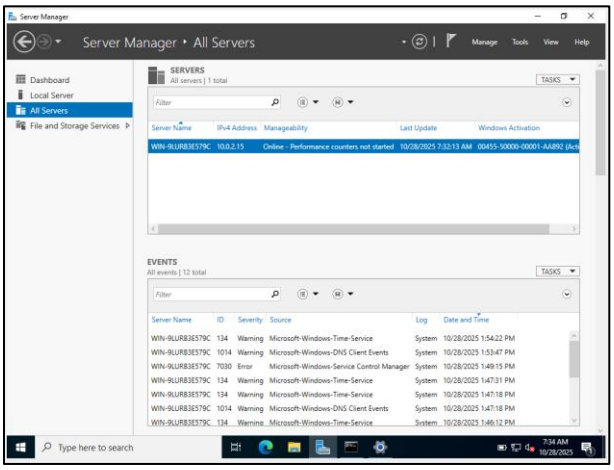
- Type ***server Manager*** in search bar & press Enter:



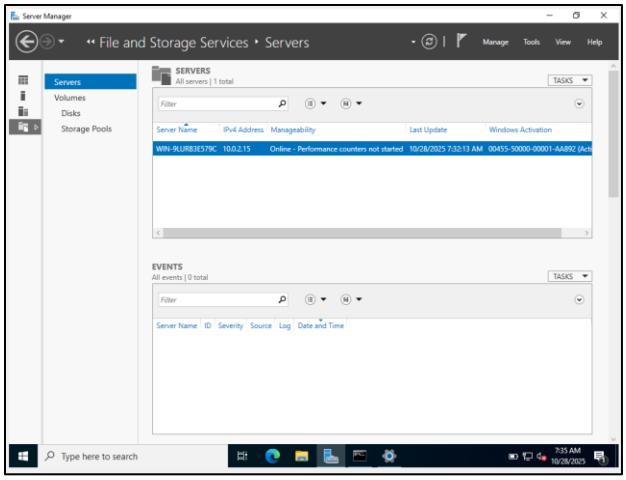
- This screen shows the details of Local servers. Move to ***second*** point (***Local Server***) below ***Dashboard*** at left upper corner.



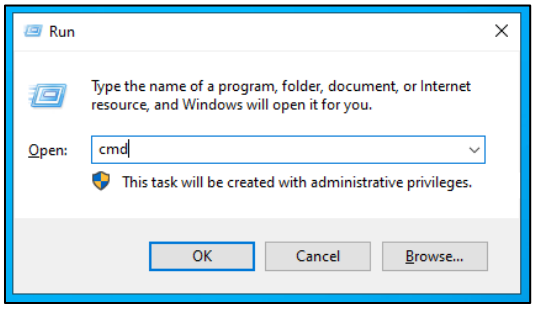
- Then move to ***Third*** point (***All Servers***) below ***Local Server***.



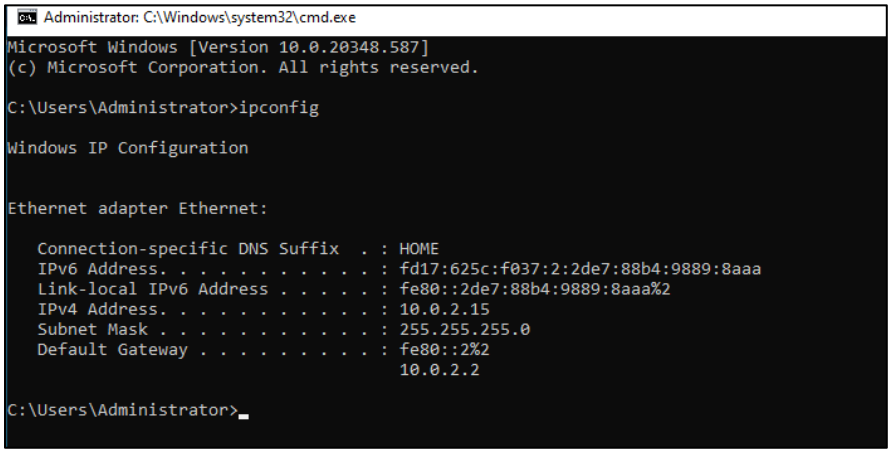
- Then move to ***Fourth*** point (***File and Storage Services***) below ***All Servers***.



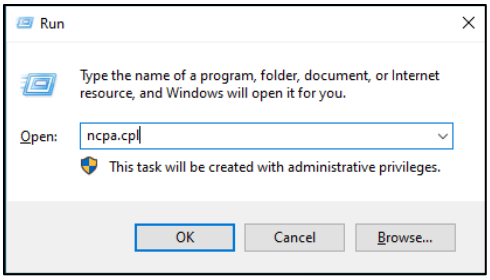
- Press ***window+R*** & ***type cmd:***



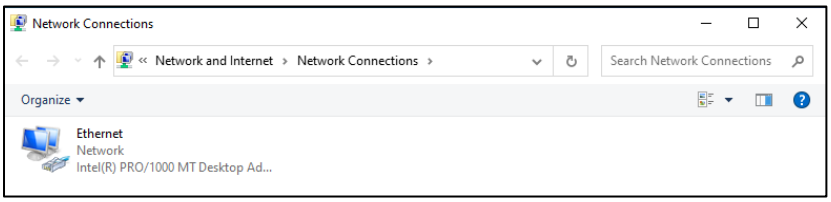
- Then type ***ipconfig*** & press ***Enter***. The screen display the windows ip configuration details.



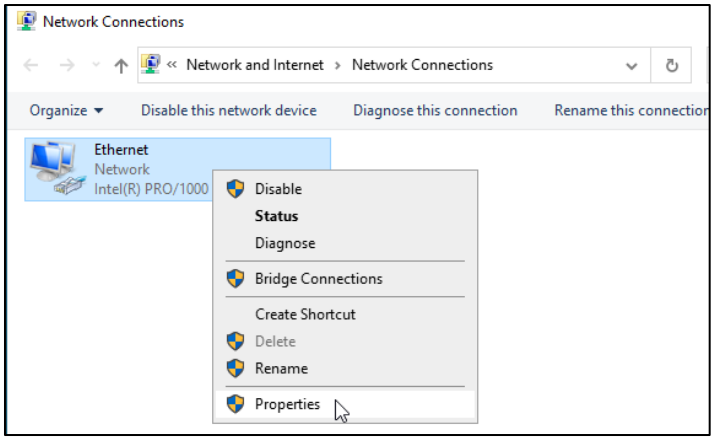
- Press ***window+R*** & ***type ncpa.cpl:***



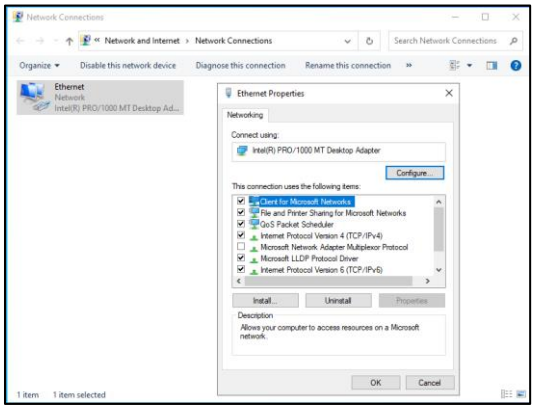
- Ethernet Network screen will be open.



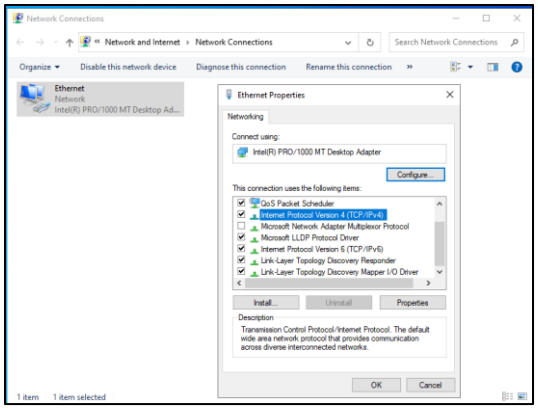
- ***Right click*** on ***Ethernet*** & choose ***last option*** (***properties***).



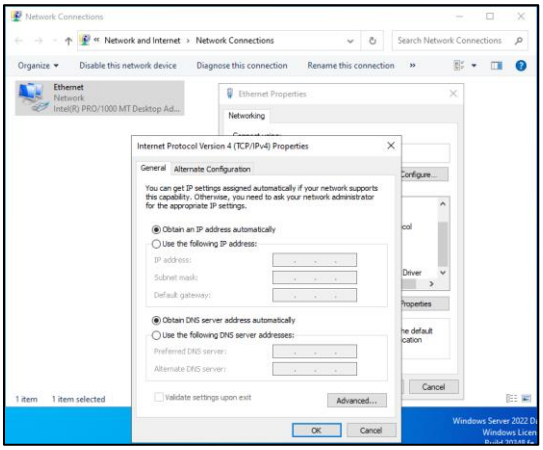
- The ***new screen*** will be open.



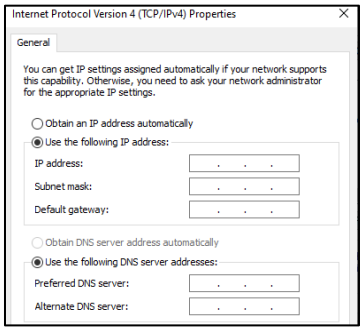
- ***Scroll down*** & double click on ***Internet Protocol Version 4(TCP/IPv4)***.



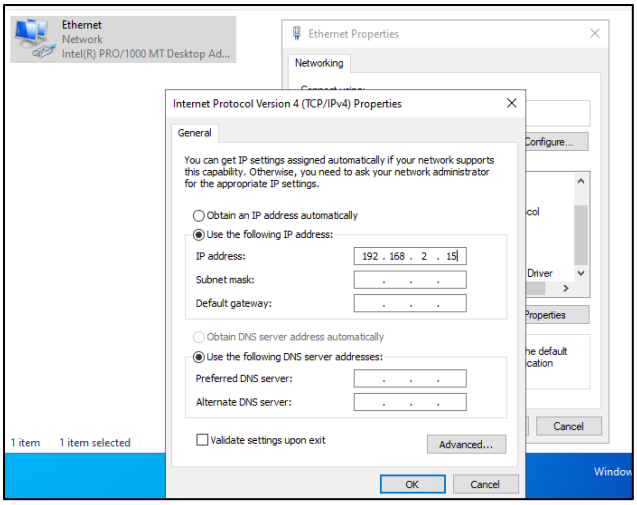
- The ***new screen*** will be open. The IP address & Preferred DNS Server address will be the one we get from ipconfig command. After adding them click on OK.



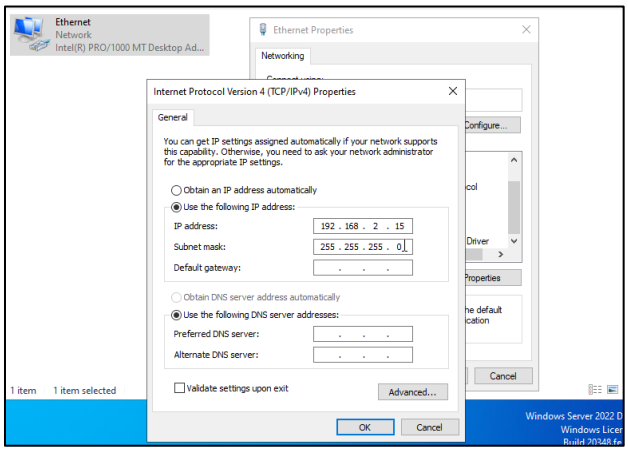
- Choose ***Use the following IP address***.



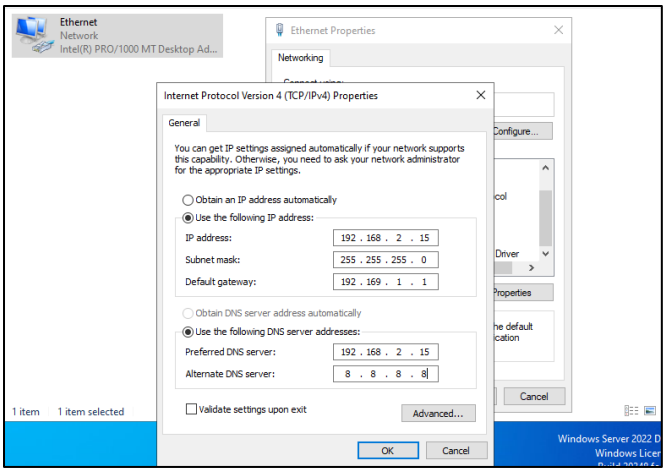
- Type ***IP address like 192.168.2.15***



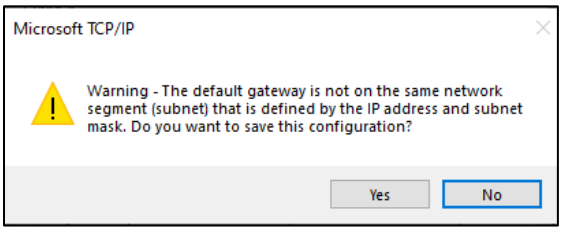
- ***Just Click on Subnet mask space***, it will be enter automatically.



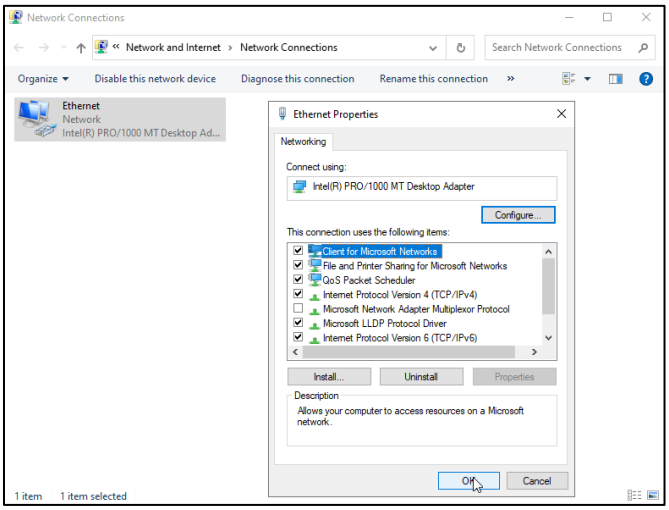
- Type ***Default gateway 192.169.1.1*** & ***Preferred DNS server*** same like IP address ***Alternate DNS server 8.8.8.8*** then click ***OK.***



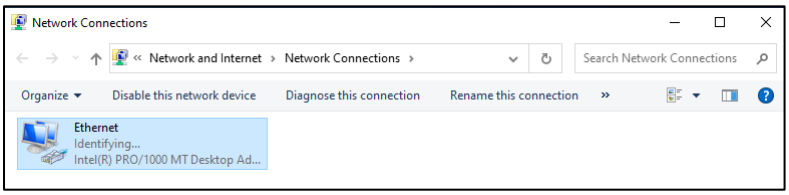
- Click on ***Yes***.



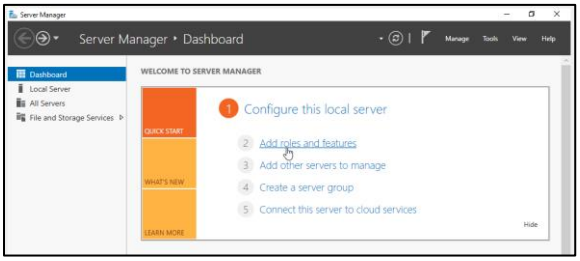
- Then click ***OK*** for previous one screen.



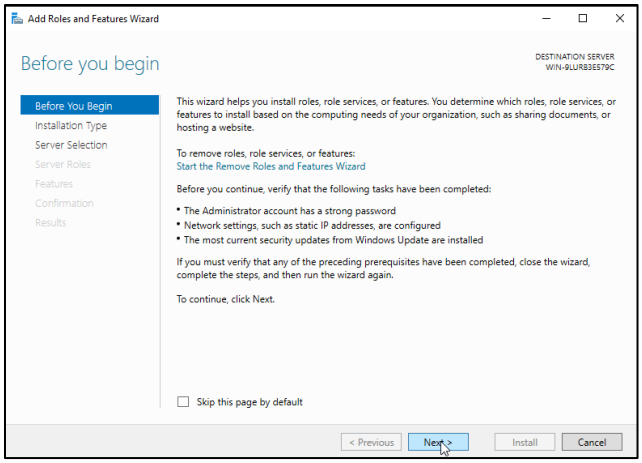
- ***Ethernet identifying Network***



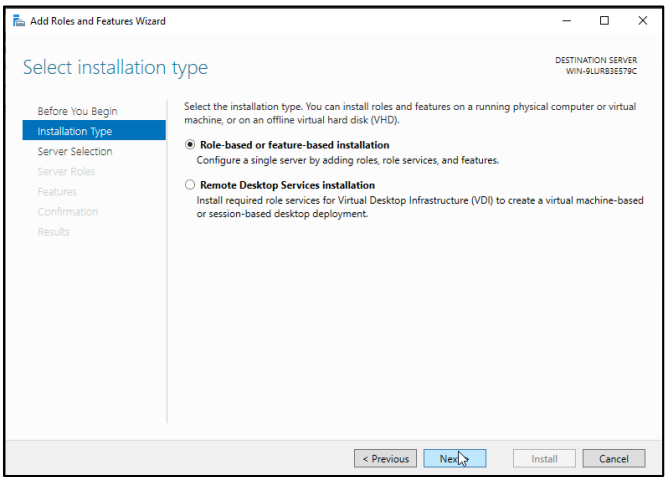
- In Server Manager screen, click on ***Add roles and features***.



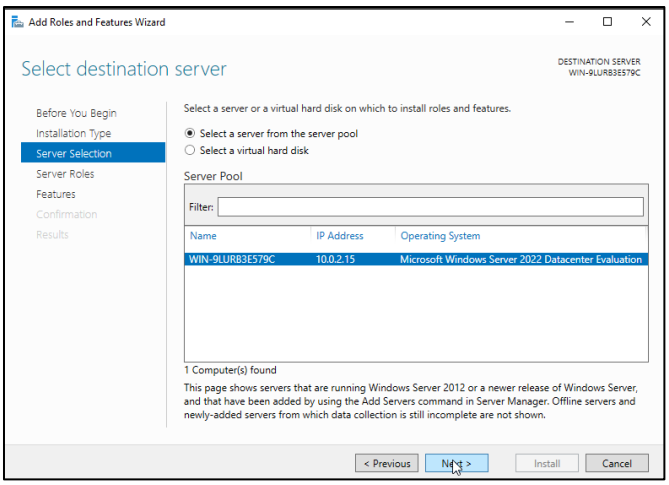
- ***Next*** for new screen.



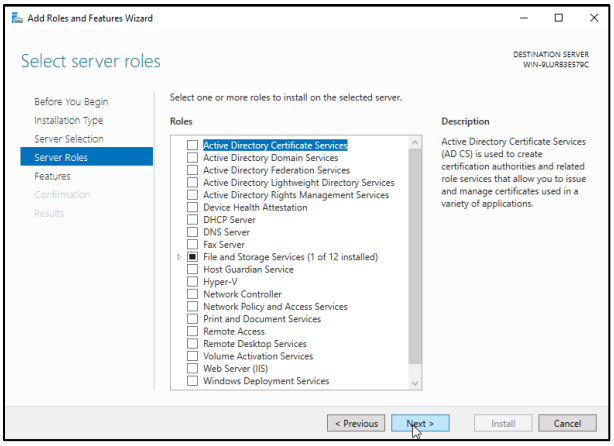
- Now the “installation type” window will appear again click on Next.



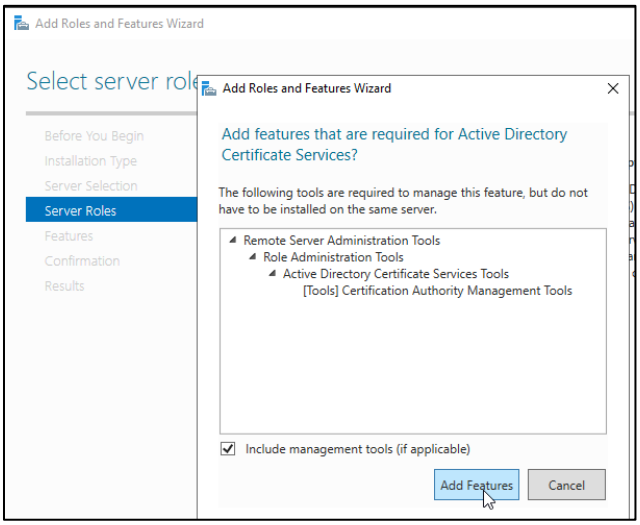
- Then The “Server Selection” Window simply click on Next to move on.



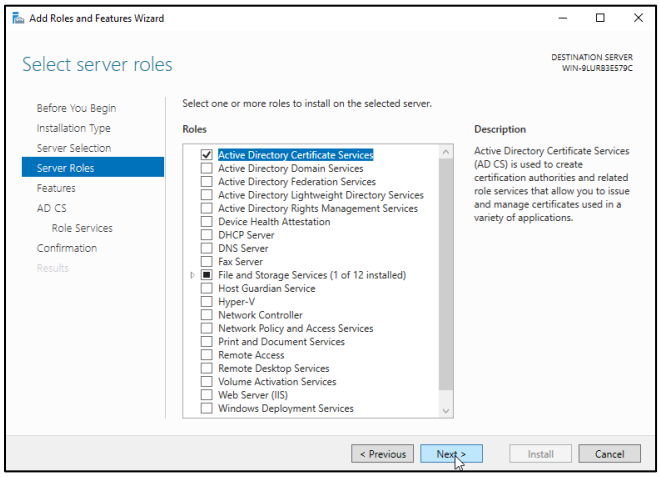
- Now click on box of ***Active Directory Certificate Services***.



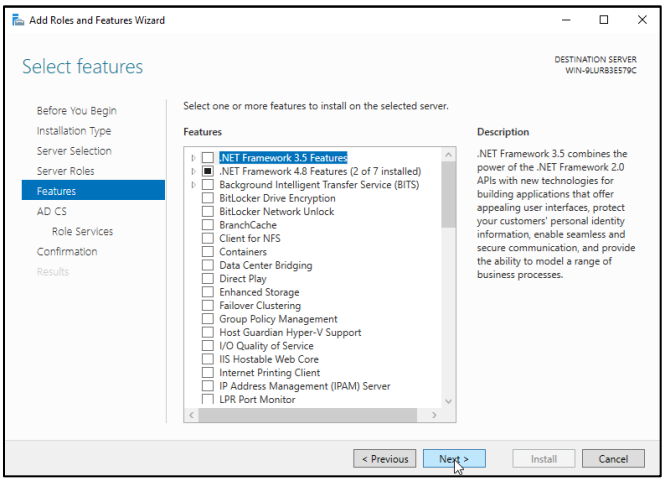
- After clicking "Active Directory Domain services" the following window appears, simply click on Add Features.



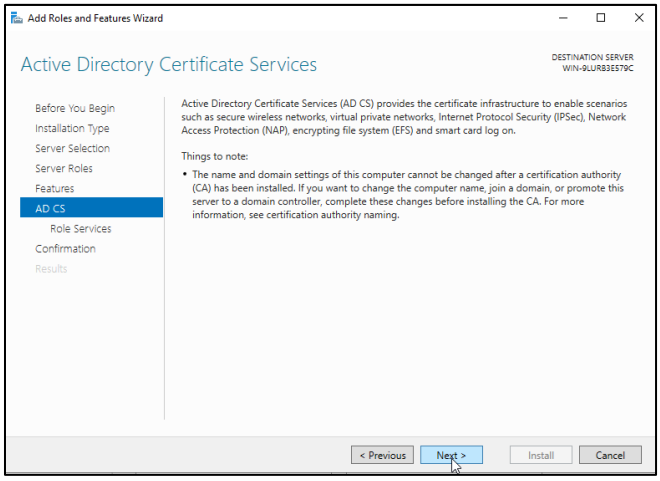
- Now "Active Directory Domain services" option is checked click on Next to move further.



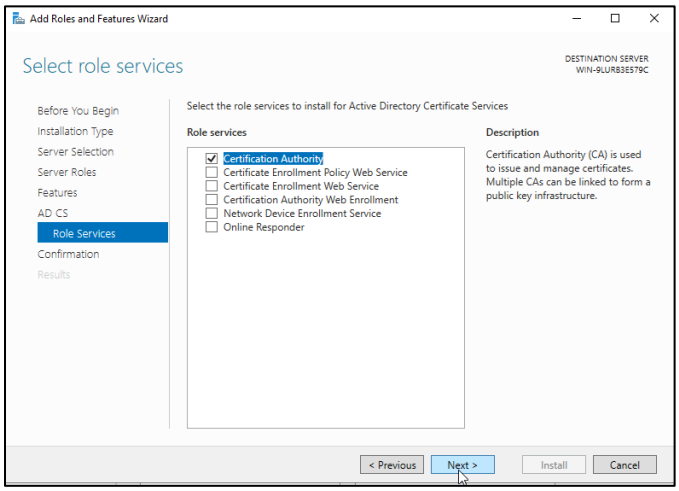
- Again ***Next***.



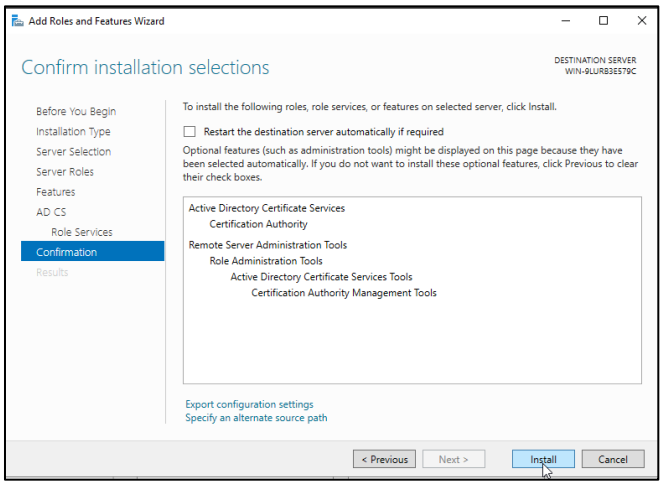
- Now "AD DS" screen appear click on Next to proceed forward.



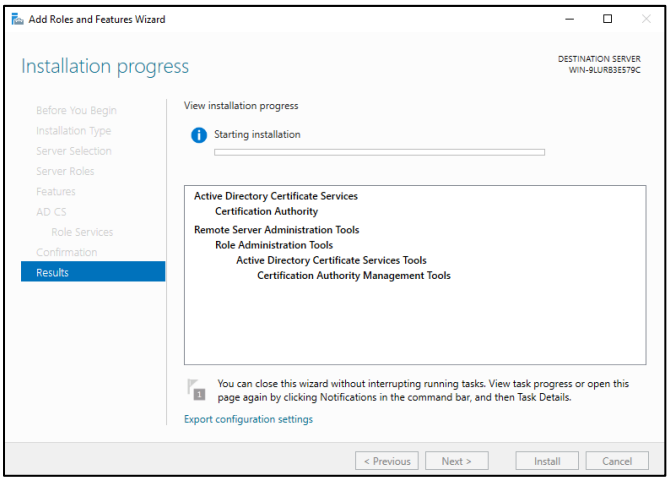
- Again ***Next***.



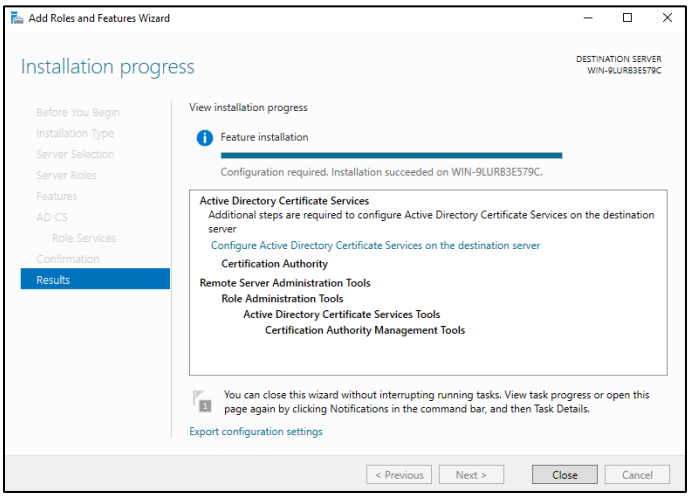
- Click ***Install***.



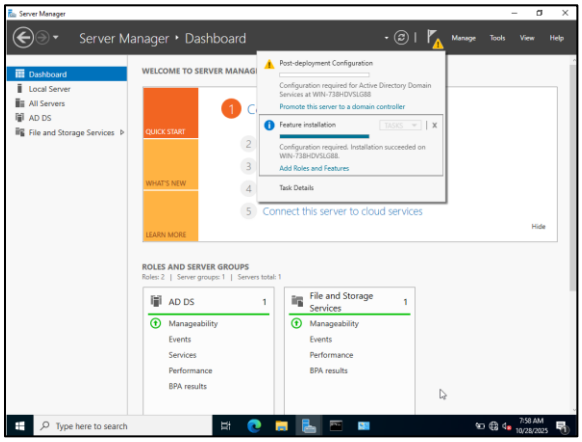
- ***Installation Start*** wait until complete.



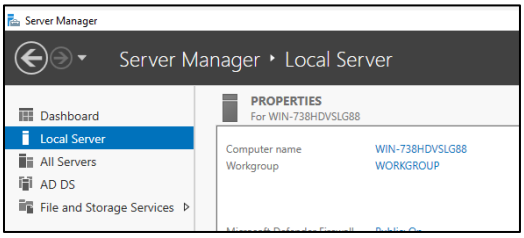
- When complete installation then press ***close***.



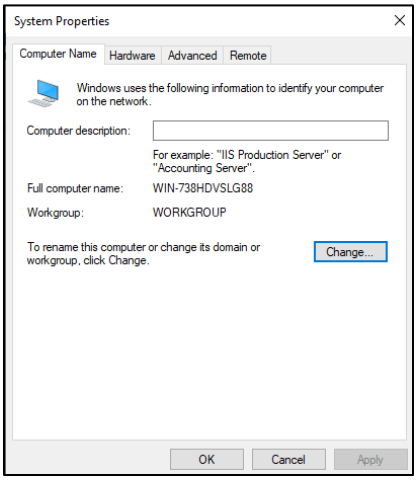
- Now Add Roles and Features process is complete.



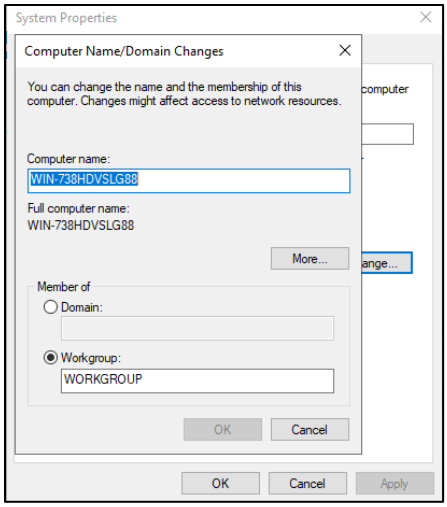
- The next step is to change the computer name. In Local server we can see the current computer name.



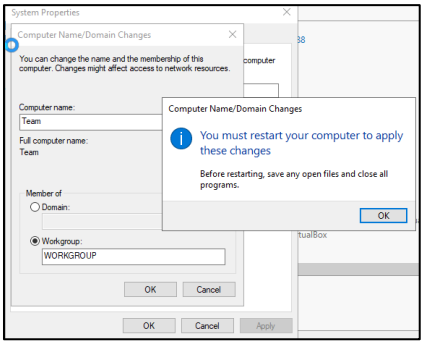
- Click on the computer name following window appear, click on change.



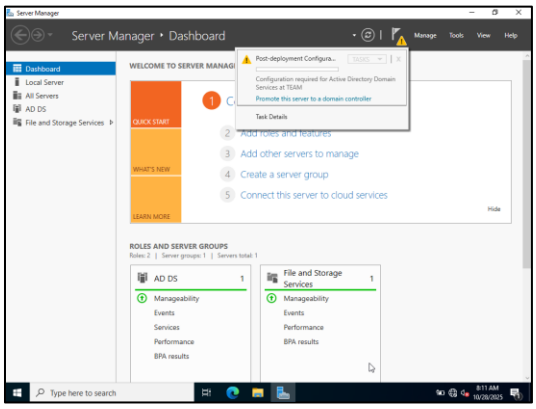
- Now this window appear change the computer name.



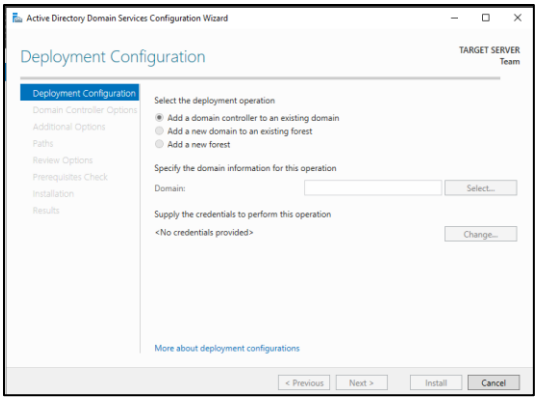
- In the following screen we can see the changed name and after changing the message of restart arise click on OK.



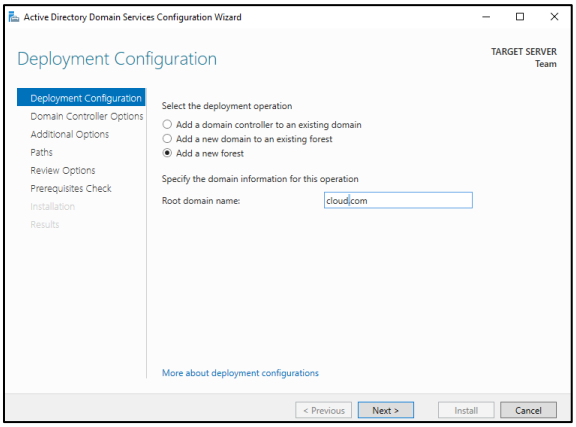
- After restart again go to server manager and click on the flag at right top corner with a warning sign. After clicking the flag following box appear click the blue text.



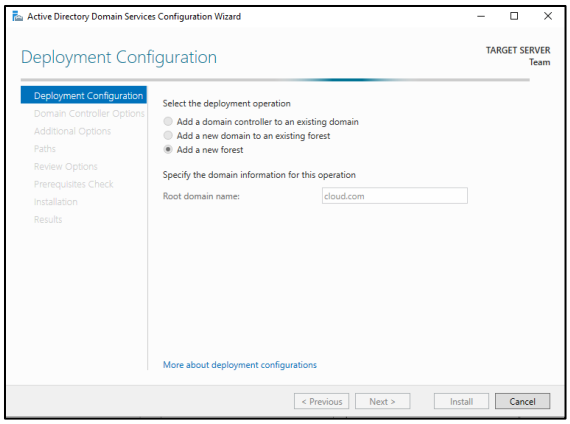
- After clicking the deployment configuration process starts.



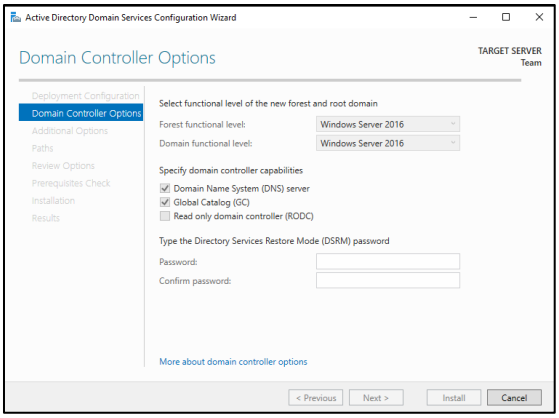
- In deployment configuration check the third option "Add a new forest". Then specify the root domain name (cloud.com) and click on Next.



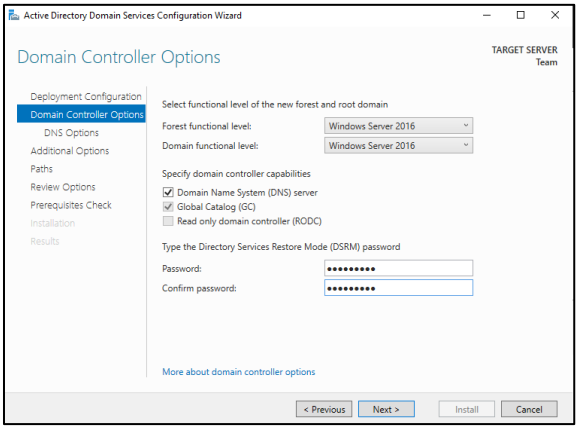
- Wait for Changes to apply.



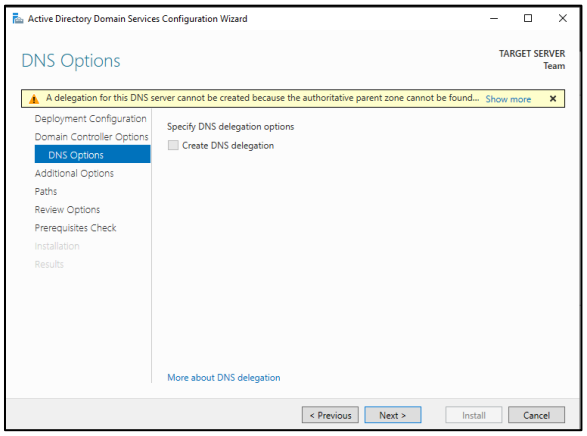
- Next step is "Domain Controller Options". In this window specify the password and confirm the password.



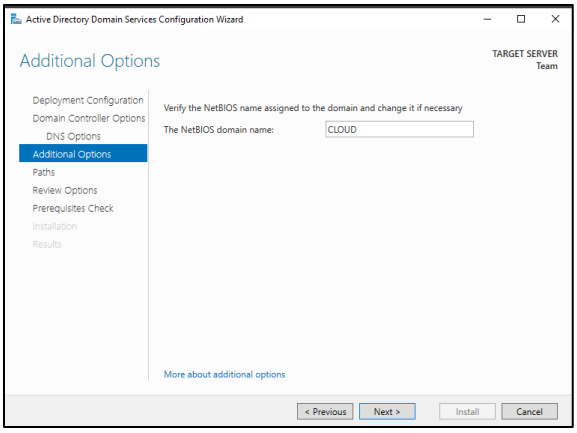
- After specifying the password click on Next.



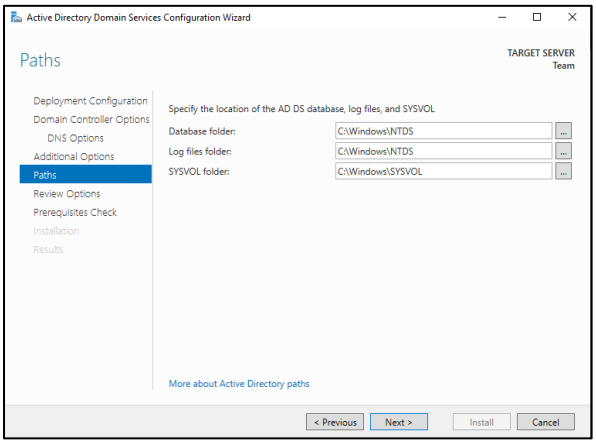
- Next is “DNS Options”. Simply click on Next.



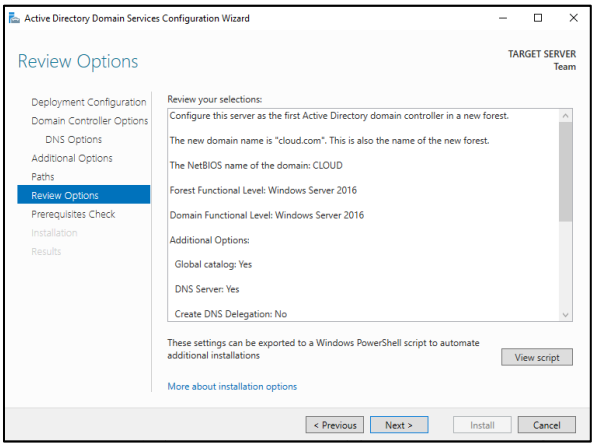
- Next window will appear in which there are additional options simply click on Next.



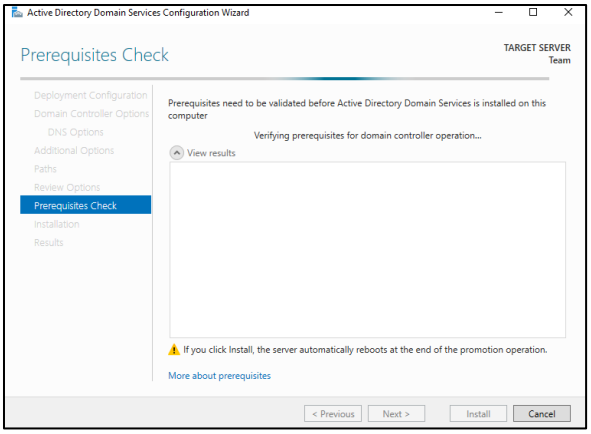
- The following window show the paths again click on Next.



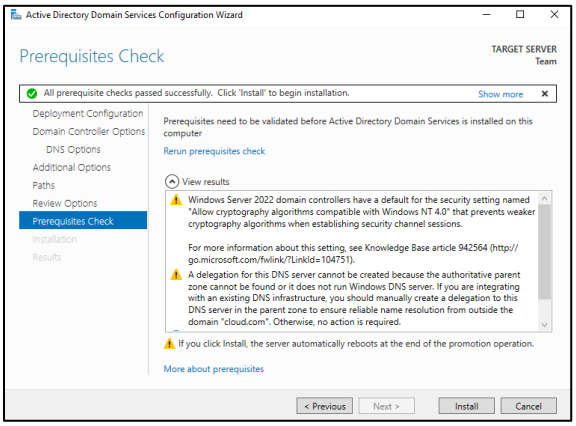
- Now “Review Option” window appears click on Next.



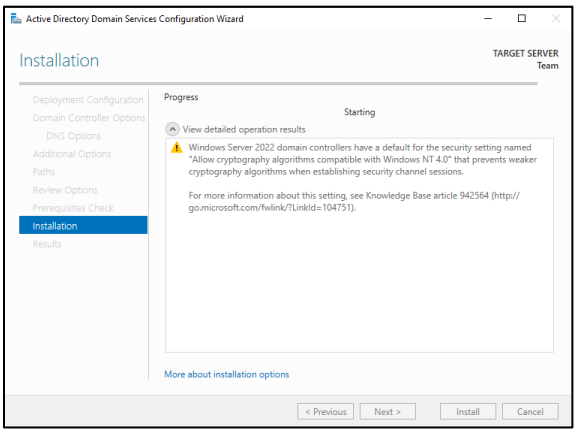
- The next screen shows Prerequisites check.



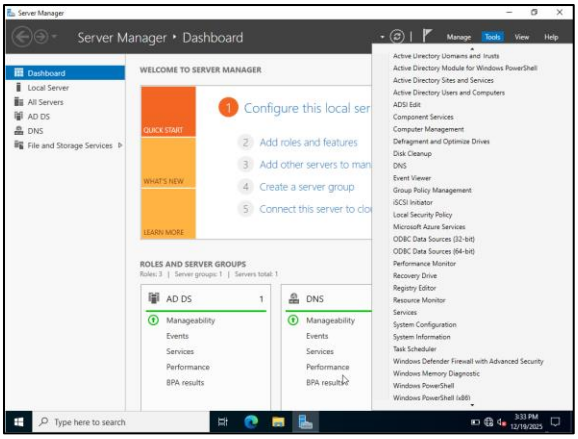
- Click on Install to process further.



- Wait for Installation to complete.

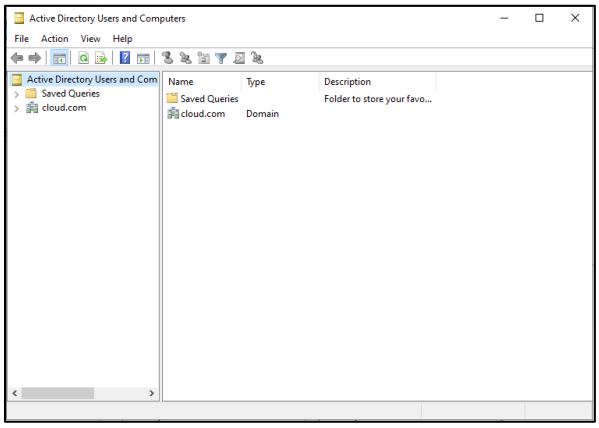


- Now the deployment configuration completes. To check the process completion click on tools and check "Active Directory users and computers" whether this option appears or not. If successful then the option will appear and we can see this option at fourth number.

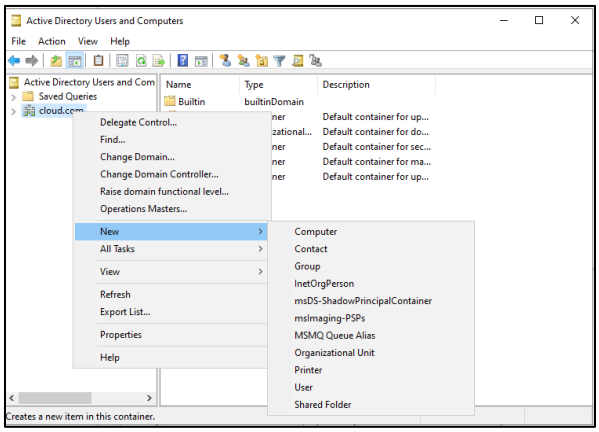


Task#1: Creating Organizational Units(OUs):

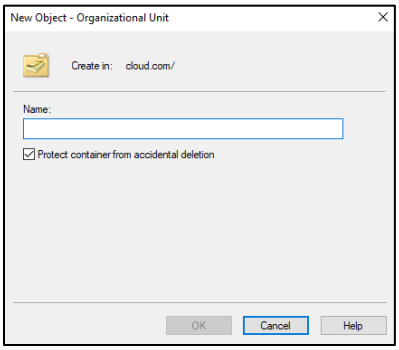
- Creating IT, HR, Finance and Sales OUs.
- Click the "Active Directory users and computers" the following screen appear having the cloud.com forest we created previously.



- Right click the cloud.com forest then click on New and then from next window Organizational Unit.

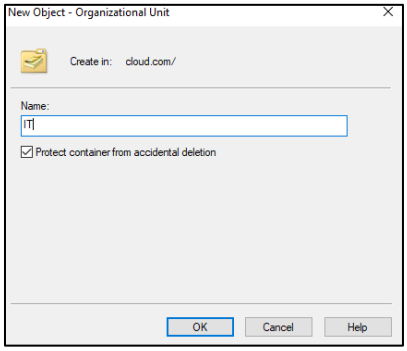


- This window will appear having name box.

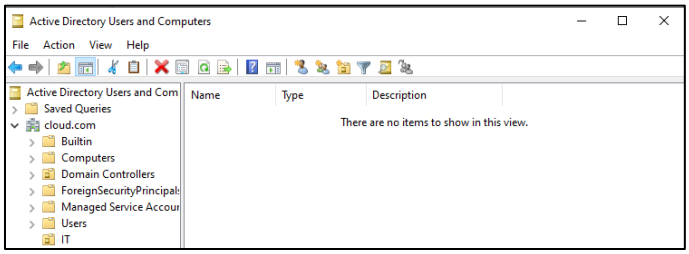


Creating IT Organizational unit:

- Type IT in name box and click OK.

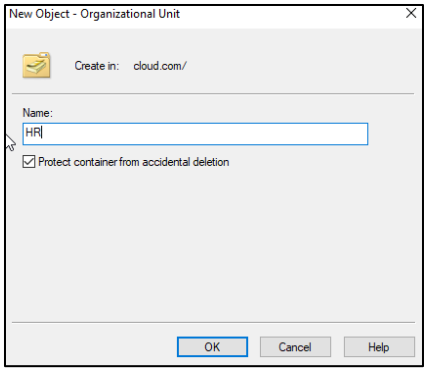


- Now we can see IT OU on the screen.

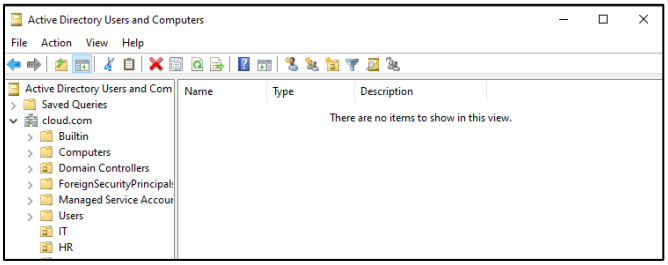


Creating HR Organizational unit:

- Again repeat the process right click cloud.com and then New and then Organizational Unit and add HR in name box and click on Ok.

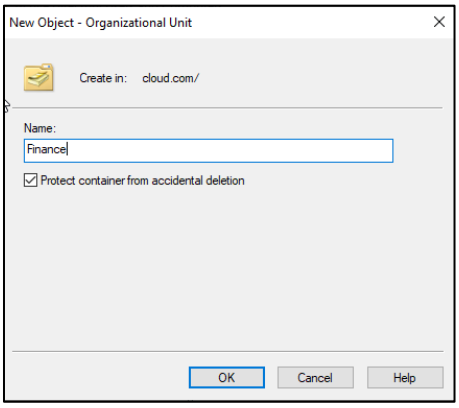


- Now we can see HR OU on the screen.

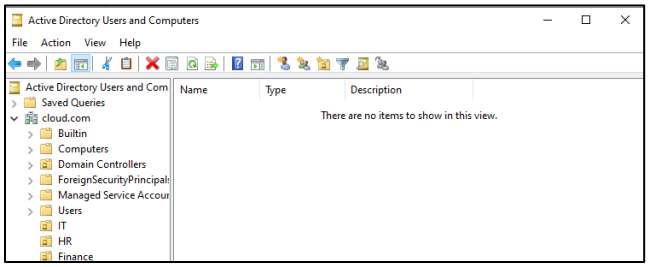


Creating Finance Organizational unit:

- Again repeat the process right click cloud.com and then New and then Organizational Unit and add Finance in name box and click on Ok.

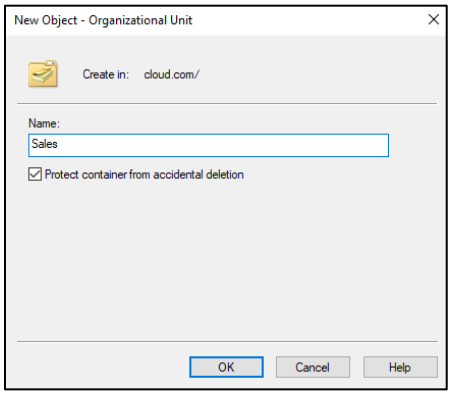


- Now we can see Finance OU on the screen.

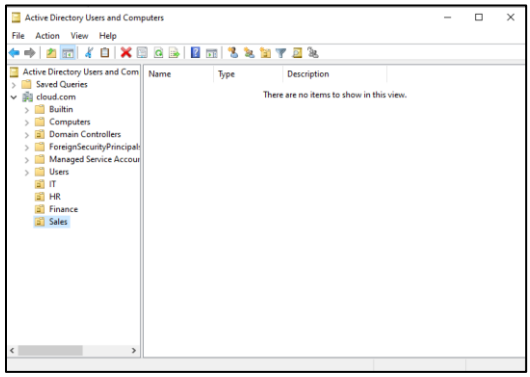


Creating Sales Organizational unit:

- Again repeat the process right click cloud.com and then New and then Organizational Unit and add Sales in name box and click on Ok

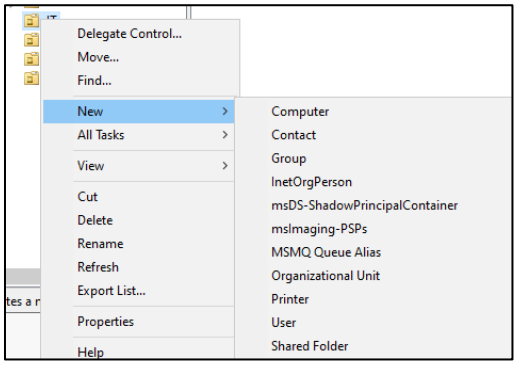


- Now we can see Sales OU along with all other OUs on the screen.

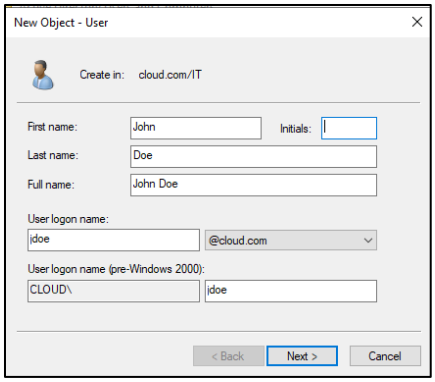


Task#2: Creating User:

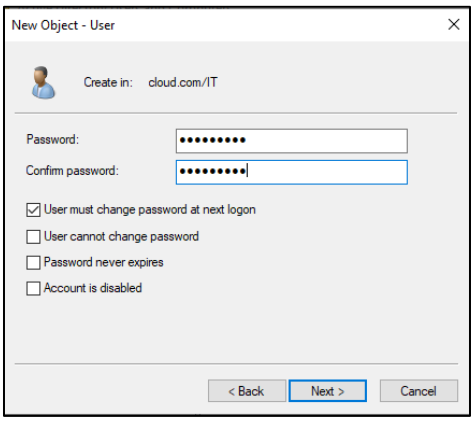
- Creating User John Doe with user name jdoe.
- For creating user right click the IT OU, then from next window click on New and then from next screen select User.



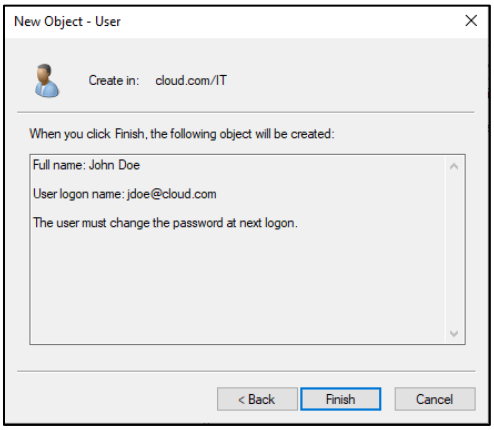
- Now this window appears enter the first name, last name, and user logon name and click on Next.



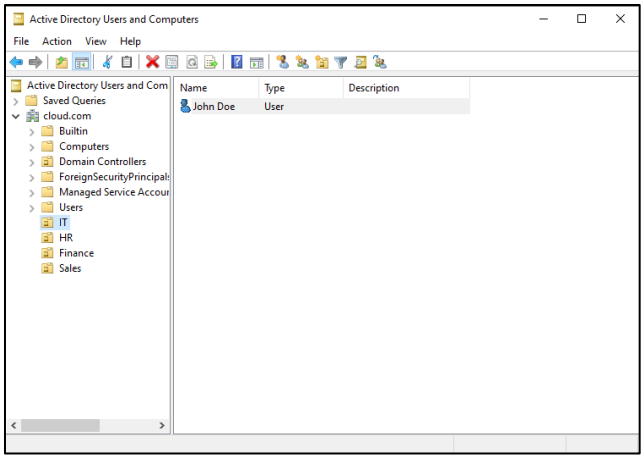
- Then type a password for user and confirm the password. Click on Next.



- Now this window appears click on finish to complete the process.

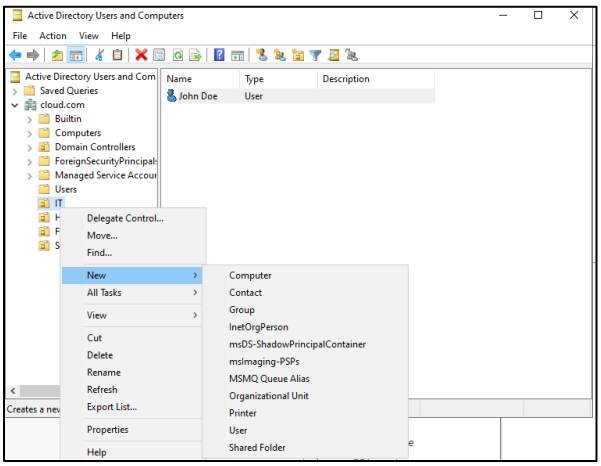


- Now we can see the user created on the screen.

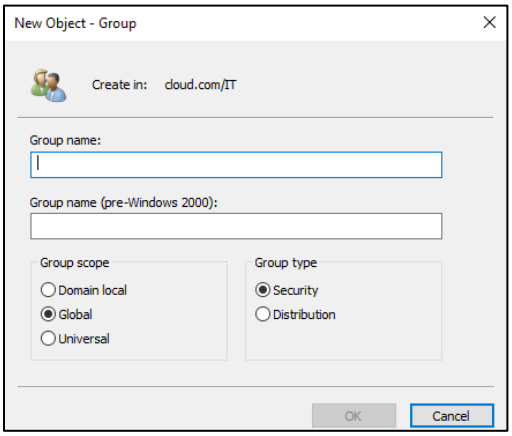


Task#3: Creating Groups:

- Creating IT_Admins and Standard_Users groups.
- For creating group again right click the IT OU and then click on New and from next window click on Group.

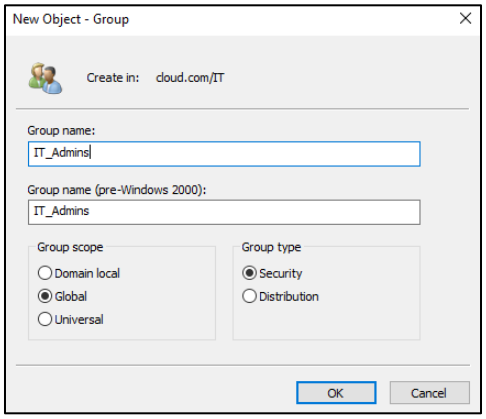


- Now this window will appear having group name.

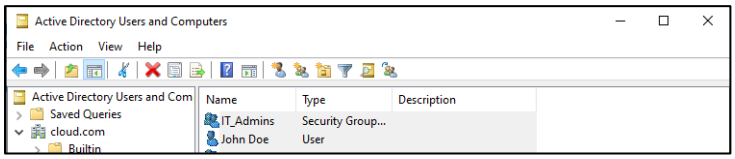


Creating IT_Admins Group:

- Enter the IT_Admins in Group name box. Click on OK.

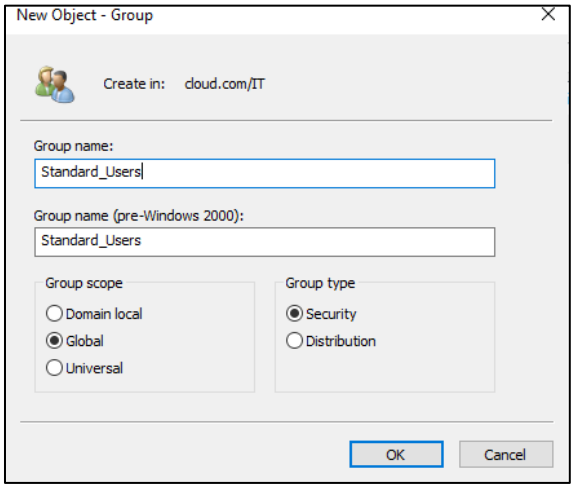


- Now we can see the IT_Admins group created on the screen.

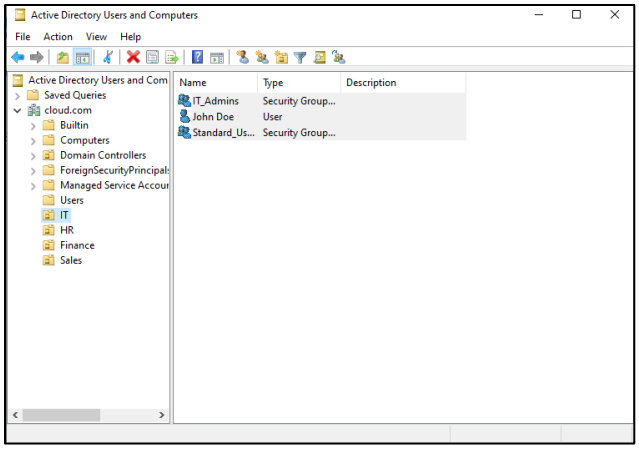


Creating Standard_Users Group:

- Enter the Standard_Users in Group name box. Click on OK.

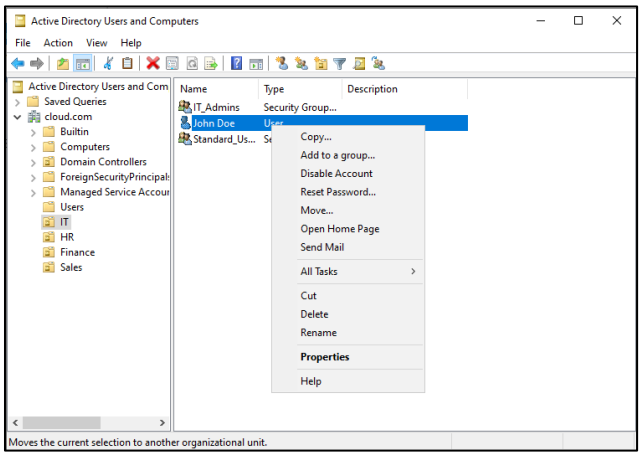


- Now we can see the Standard_Users group created on the screen.

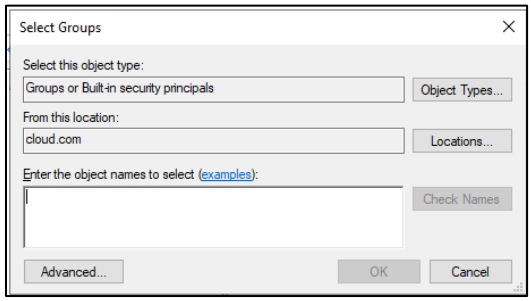


Adding User to Groups:

- Right click the JohnDoe User then click on Add to group.

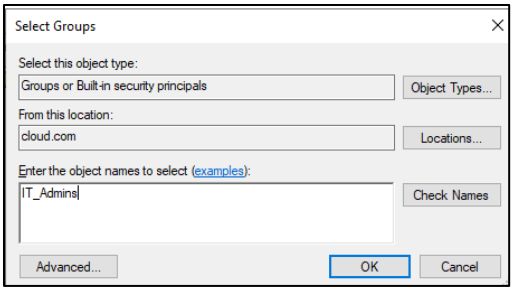


- Now the following screen appear in which we enter the group name in which we want add user.

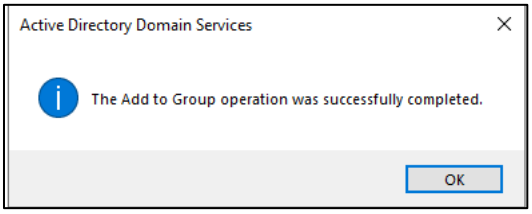


Adding JohnDoe User to IT_Admins Group:

- Now add the group name IT_Admins in the box and click OK.

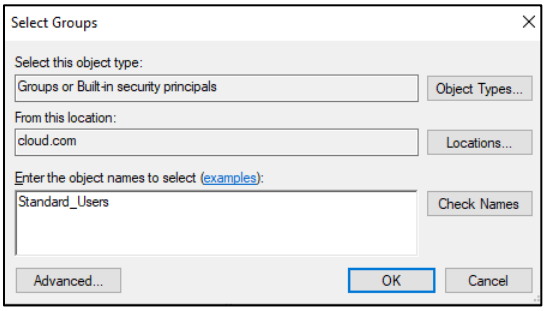


- Operation completed successfully.

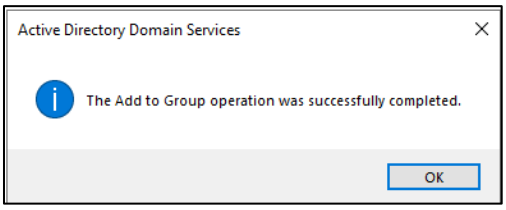


Adding JohnDoe User to Standard_Users Group:

- Now add the group name Standard_Users in the box and click OK.

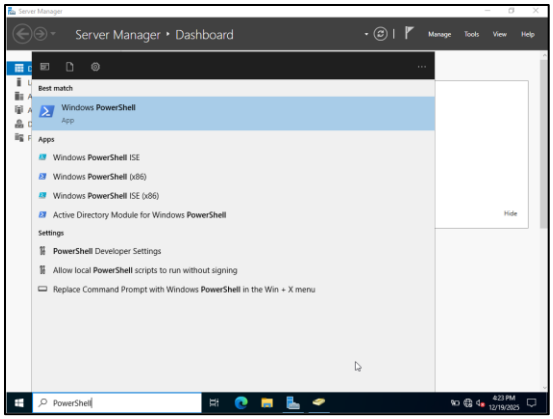


- Operation completed successfully.

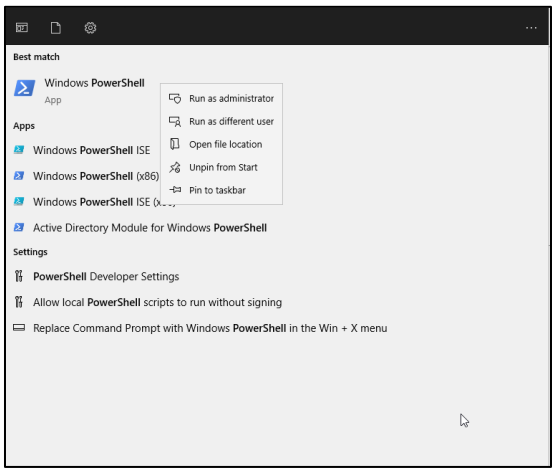


Bulk Create Users & Groups with PowerShell

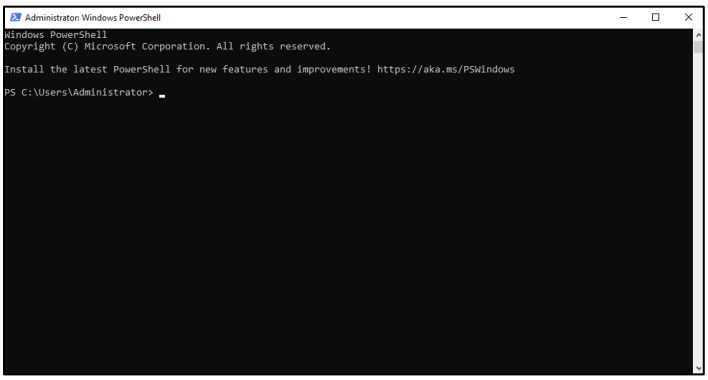
- For adding multiple users and groups at once instead of doing individually open the window powershell.



- Run it as administrator.



- This screen will appear.

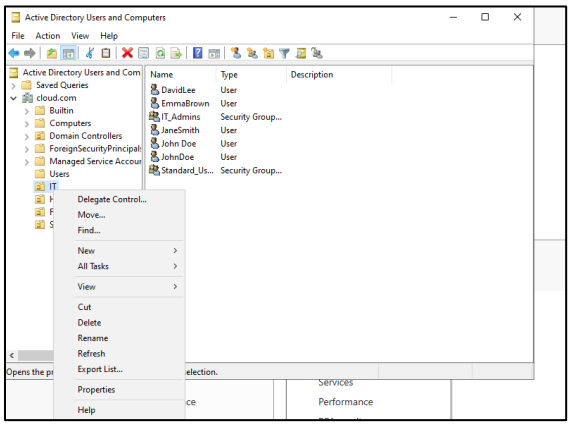


- Type the following command in correct syntax and press enter.

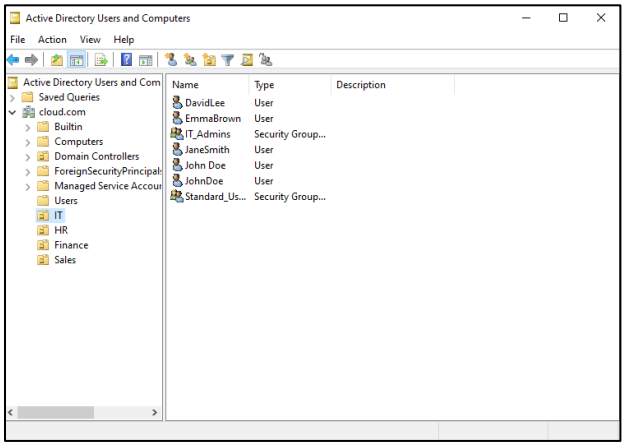
$Users = @("JohnDoe","JaneSmith","DavidLee","EmmaBrown") ;foreach ($User in $Users) { New-AD-User -Name $User -SamAccountName $User -UserPrincipalName "$User@techgenius.com" -Path "OU=IT,DC=cloud,DC=com" -AccountPassword (ConvertTo-SecureString "cloud@2025!" -AsPlainText -Force) -Enabled $true }

```
PS C:\Users\Administrator> $Users = @("JohnDoe","JaneSmith","DavidLee","EmmaBrown") ; foreach ($User in $Users) {
>> New-AdUser -Name  $User -SamAccountName $User -UserPrincipalName "$User@techgenius.com" -Path "OU=IT,DC=cloud,DC=com"
-AccountPassword(ConvertTo-SecureString "cloud@2025!" -AsPlainText -Force) -Enabled $true }
PS C:\Users\Administrator>
```

- Now go back to the Active directory users and groups screen. Right click IT OU and Refresh it by clicking Refresh.



- Now we can see all users on the screen. Command run successfully.



THE END!